MIL-R-42016(DM)
2 December 1986

MILITARY SPECIFICATION

RONGEUR, Straight, Square Bite, 30.5 cm Long, (Sauerbruch)

This specification is approved for use within the Defense Personnel Support Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rongeur of the Sauerbruch design.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the exist specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping Fiberboard.
PPP-B-676	- Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

24535 - RONGEUR, SAUERBRUCH, 17 mm Wide Jaw

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The rongeur shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The rongeur shall be fabricated from Class 4 martensitic stainless steel in accordance with ASTM F899.

3.3 Hardness. The Rockwell hardness of the cutting surfaces shall be C50 to C55 when tested as specified in 4.4.1. The remainder of the instrument shall be C45 to C50. The requirements of GG-I-526 shall apply.

3.4 Style, design and dimensions. Shall be a straight rongeur of the Sauerbruch design, used to take square bites of ribs during surgical procedures. The rongeur shall have a double action mechanism, cupped jaws, and an overall length of 30.5 cm. The handle shall have a hand stop to prevent slippage during use. When closed, the cutting surfaces shall mate completely. The action of the rongeur shall be easy, without drag or binding. The rongeur shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24535.

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3.5 Mechanism. Construction shall be accurately fitted, and shall be without crevices, burrs, or sharp edges at any place in its construction. The joint performance shall be smooth, of equal resistance and non-binding in all positions when opening or closing. Further, under no circumstances shall looseness or free play of either shank permit any misalignment of the jaws in closing. NOTE: This paragraph shall replace the requirements specified in GG-I-526 for locks.

3.6 Finish. Finish shall be entirely mirror, satin or matte. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.6.1 Passivation. The rongeur shall be properly passivated.

3.8 Identification markings. In addition to the requirements specified in GG-I-526, instruments of foreign manufacture shall be marked "stainless."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.3 Certificate of quality. Certificates of quality supplied by the manufacturer of the corrosion-resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purposes shall be one rongeur.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one rongeur. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
Dimensions	3.4	---	S-2	2.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the rongeur shall be selected for chemical analysis.

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4.3 Examination.

4.3.1 Examination of rongeur. The rongeur shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any rongeur in the sample containing one or more defects shall be rejected and if the number of defective rongeurs in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Rongeurs not of the specified design for appropriate type, stock number and item identification.
102	Rongeurs not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
103	Mechanism performance not as specified in 3.5.
Minor	
201	Rongeurs not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5 Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Rongeurs shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each rongeur shall be packaged in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.4 Intermediate package. Six unit packages shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable specification box.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for rongeur shall conform to the MIL-STD-794 requirements for level C.

5.1.2.2 Intermediate package. Six intermediate packages shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547, Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Eight intermediate packages (48 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Eight intermediate packages (48 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Rongeurs packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Rongeurs packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The Sauerbruch rongeur shall be used to take square bites of ribs during surgical procedures.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number Coverage. This specification is intended to cover the following National Stock Number (NSN):

National Stock No.	Item Identification
6515-01-139-8115	RONGEUR, SAUERBRUCH, 30.5 cm Long, Double Action, Straight, 17 mm Wide Jaw, Square Bite

6.4 Matte and satin finish. It should be noted that the matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistant then satin finish.

6.5 Subject term (key word) listing.

Rib rongeur
Rongeur
Sauerbruch rongeur

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3852

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-42016(DM)	RONGEUR, Straight, Square Bite, 30.5cm Long, (Sauerbruch

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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